Software Testing Workshop 01/06/2022

1. Functionality Tests

1.1. Arithmetic Operation Tests

1.1.1. Addition Test

1.1.1.1. Case 1

Aim: Do verification for addition operation.

Inputs: 1453+2022+2001

Steps: A number entered then + sign pressed and went on like that until reach at end of the input list.

Expected Output: 5476

Given Output: 4023

Result: Fail

1.1.2. Subtraction Test

1.1.2.1. Case 1

Aim: Do verification for subtraction operation.

Inputs: 2022-2001

Steps: A number entered then + sign pressed and went on like that until reach at end of the input list.

Expected Output: 21

Given Output: 21

Result: Success

1.1.3. Multiplication Test

1.1.3.1. Case 1

Aim: Do verification for multiplication operation.

Inputs: 8*8

Steps: A number entered then * sign pressed and went on like that until reach at end of the input list.

Expected Output: 64

Given Output: 64

Result: Success

1.1.3.2. Case 2

Aim: Do verification for multiplication operation.

Inputs: -8*-8

Steps: A number entered then * sign pressed and went on like that until reach at end of the input list.

Expected Output: 64

Given Output: 8

Result: Fail

1.1.4. Division Test

1.1.4.1. Case 1

Aim: Do verification for division operation.

Inputs: 86/9

Steps: A number entered then / sign pressed and went on like that until reach at end of the input list.

Expected Output: 9.55

Given Output: 9.55

Result: Success

1.1.5. Logarithm Test

1.1.5.1. Case 1

Aim: Do verification for logarithm operation.

Inputs: 128log

Steps: A number entered then log sign pressed and went on like that until reach at end of the input list.

Expected Output: 2.107

Given Output: 2.107

Result: Success

1.1.6. Square Root Test

1.1.6.1. Case 1

Aim: Do verification for square root operation.

Inputs: 16 Sqrt

Steps: A number entered then Sqrt sign pressed and went on like that until reach at end of the input list.

Expected Output: 4

Given Output: 4

Result: Success

1.2. Trigonometric Function Tests

1.2.1. Sine Test

1.2.1.1. Case 1

Aim: Do verification for sine operation.

Inputs: -60 sin

Steps: A number entered then sin sign pressed and went on like that until reach at end of the input list.

Expected Output: -0.86

Given Output: -0.86

Result: Success

1.2.2. Cosine Test

1.2.2.1. Case 1

Aim: Do verification for cosine operation.

Inputs: -397 cos

Steps: A number entered then cos sign pressed and went on like that until reach at end of the input list.

Expected Output: 0.7986

Given Output: 0.7986

Result: Success

1.2.3. Tangent Test

1.2.3.1. Case 1

Aim: Do verification for tangent operation.

Inputs: 64 tan

Steps: A number entered then tan sign pressed and went on like that until reach at end of the input list.

Expected Output: 2.05

Given Output: 2.05

Result: Success

1.3. Mixed Tests

1.3.1. Case 1

Aim: Do verification for a combined trigonometric operation.

Inputs: 60 sin / 60 cos

Steps: Given input entered to the calculator.

Expected Output: 1.73

Given Output: 1.73

Result: Success

1.3.2. Case 2

Aim: Do verification for the exponent operation.

Inputs: 8 e (e^8)

Steps: Given input entered to the calculator.

Expected Output: 2980.95

Given Output: 2980.95

Result: Success

1.3.3. Case 3

Aim: Do verification for the 1/x operation.

Inputs: 1/x

Steps: Given input entered to the calculator.

Expected Output: NaN

Given Output: Infinity

Result: Fail

1.3.4.

2. Bugs and Defects

2.1. Bug 1

Definition: The calculator can only do operations with entered the last two numbers. It doesn't work if more than two operands entered.

Severity: Major

2.2. Bug 2

Definition: If both entered numbers are negative it gives a wrong result.

Severity: Major

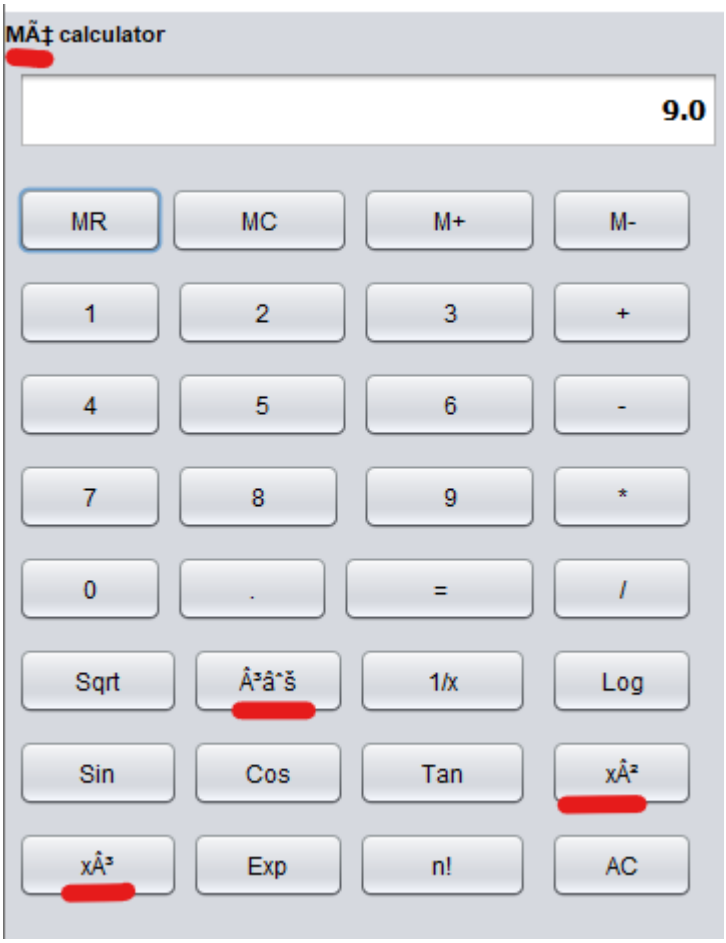
2.3. Bug 3

Definition: Trigonometric functions works with degree only. It does not support radian.

Severity: Minor

2.4. Bug 4

Definition: There are some unrecognizable strings on GUI. The buttons are operational, but we don't know what they do.



Severity: Major

2.5. Bug 5

Definition: 1/x should be NaN when zero entered.

Severity: Major

2.6.